MLFF Training-Data and Fine-Tuning Architecture

Purpose and authority

This manual defines the accepted current scientific, statistical, execution, and evidence architecture for the mdstats MLFF workflow: source-certified atomistic data preparation, leakage-safe evidence roles, neutral statistical preparation, one target-size screen, MACE fine-tuning, selected-only method validation, fresh final production, and bounded campaign execution. Downstream deployment, physical, calibration, and locked-test capabilities remain separately owned product obligations.

It is intentionally present-tense and single-generation. A reader does not need release chronology, migration history, or obsolete stage semantics to determine current behavior.

The canonical editable architecture sources are the numbered chapters under `docs/arch_manuals/mlff_training_data/`. The assembled Markdown and PDF are generated publication products of those sources and must not be edited as independent authorities.

Detailed exact behavior is owned by current specifications under `docs/specs/training_data/`. Methods/theory material may explain rationale but does not override architecture or specifications. Proposed transitions live in `workplans/`; completed chronology lives under `docs/history/mlff/`; correctness/performance evidence lives in audits, release evidence, and benchmarks.

Architectural motive

MLFF campaigns combine state with fundamentally different epistemic roles: physical source facts, eligibility decisions, evidence partitions, fitted transforms, subset-membership decisions, target-size decisions, optimization/checkpoint state, protocol-validation evidence, calibration evidence, locked tests, and deployment decisions. Conflating those roles creates leakage and ambiguous authority even when the numerical code is correct.

The architecture therefore uses immutable/content-addressed evidence, explicit statistical roles, one normative owner per scientific decision, and authenticated dependency direction. Execution realization is kept separate: cache layout, worker count, queue order, out-of-core storage, and scheduler policy may change without changing scientific membership, ordering, coverage, ranking, or evidence roles.

Expensive exact numerical work is computed once per semantic identity and reused wherever its inputs are unchanged. Exactness, deterministic authoritative decisions, bounded materialization, explicit resource ownership, and restartable authenticated state take precedence over nominal utilization.

Current workflow at a glance

```
source evidence and labels
  -> eligibility / physical conditions
  -> raw feature and event evidence
  -> evidence-role partitioning
  -> neutral statistical substrate and protected relations
  -> fitted descriptors, metrics, E0/objective/weight inputs
  -> one P_train / M3 target-size development split
  -> one canonical training order pi_train and evaluation ladder M1 subset M2 subset M3
  -> one common deterministic target-size preparation
  -> paired optimizer-seed screen over candidate sizes
  -> one target-size reducer
  -> N_selected and T_selected = pi_train[:N_selected]
  -> post-selection cross-validation on exactly T_selected
  -> fresh final production on the complete T_selected
  -> currentness-fenced final-production publication
```

The current graph has exactly one target-size architecture. The retired per-domain multi-view selection generation is not an alternate current path: it is neither migrated nor semantically read forward, and a workspace still holding its derived state is rejected with an actionable destructive reset/reprepare requirement before any candidate, checkpoint, or descendant is reused. Raw scientific inputs and independently valid low-level content caches remain reusable when their recipes do not depend on retired target-size semantics.

Reading index

Need	Primary chapter
Scientific motivation, record/evidence model, and scope	Part I - Foundations
Source identity, labels, strain/stress, eligibility, raw features/events	Part II - Data and evidence contracts
Evidence roles, leakage-safe CV, fitted preparation, objective/weighting/exposure boundaries	Part III - Statistical design and fitted preparation
Replay, MACE protocol, checkpointing, validation, deployment, calibration, active learning	Part IV - Training, evaluation, and deployment
Target-size split/orders, paired-seed screen, reducer, post-selection CV, fresh final production	Part V - Target-size selection and post-selection validation
Exact execution, bounded resource/materialization, cache/restart/storage/progress	Part VI - Performance and execution architecture
Sole-owner matrix and accepted extension boundaries	Part VII - Ownership and extension boundaries
External scientific/algorithmic sources	References

Context retrieval index

For targeted human or AI loading, use the smallest current source containing the needed concept:

Query terms	Load first
source/label identity, eligibility, strain/stress, raw features/events	`20_data_contracts.md`
evidence roles, leakage, CV, fitted metrics, E0, objective, weighting, exposure	`30_statistical_design.md`
replay, MACE, checkpoint, evaluation, deployment, calibration, active learning	`40_training_evaluation.md`
target size, `pi_train`, `T_selected`, `M1/M2/M3`, `n1/n2/n3`, post-selection CV, final production	`50_target_size_selection.md`
scheduler, sparse execution, out-of-core, memory, persistence, progress	`60_execution_performance.md`
owner, dependency direction, unsupported generation, extension boundary	`80_ownership_and_decisions.md`
scientific/algorithmic provenance	`90_references.md`
superseded design rationale or release chronology	`docs/history/mlff/`
proposed transition	`workplans/active/`

Stable terminology

- **training domain** — an authorized gradient-training evidence partition. The current target-size choice is global; post-selection CV may derive fold-local partitions only inside `T_selected`.
- **target membership** — frame membership in a target-training subset; an exact prefix of the one canonical training order `pi_train`.
- **target size** — the protocol-level scientific target-training cardinality chosen by the one target-size reducer.
- **monitor size** — the cardinality of a monitoring/evaluation evidence set; never target-size authority.
- **training order** — the one canonical deterministic ordering `pi_train` of the target-training pool whose prefixes define candidate target subsets.
- **qualified size** — a candidate size admitted by the configured target-size policy for the current experiment definition.
- **selected size** — the one target size `N_selected` frozen by the reducer together with the exact membership `T_selected`.
- **authoritative evidence** — persisted information that defines or independently proves a scientific decision.
- **reconstructible execution cache** — discardable state derivable exactly from authoritative inputs.
- **unsupported generation** — an old campaign/artifact generation that current architecture does not interpret or migrate; it requires re-preparation.

Normative vocabulary

- **SHALL / MUST** — required for scientific, statistical, or execution correctness.
- **SHOULD** — the default design unless measured evidence justifies another exact-equivalent realization.
- **MAY** — optional realization that cannot weaken the scientific contract.

When architecture explains a change-sensitive constant whose exact value is specification-owned, the owning specification remains the sole normative location for changing that value.

Retrieval and local-context rule

Each major chapter states what its concepts own, consume, emit, and explicitly do not own. Equations and symbols are defined near first use. A chapter may repeat a dependency boundary for local comprehension, but repeated prose must not create a second independently tunable contract.

Part I - Foundations

What an MLFF learns

An energy-conserving machine-learned force field represents a potential-energy function

$$E_\theta = E_\theta(\mathbf{Z}, \mathbf{R}, \mathbf{H}),$$

where $\mathbf{Z}$ contains atomic numbers, $\mathbf{R}$ positions, $\mathbf{H}$ the periodic cell, and θ model parameters. Forces and stress follow from derivatives of the same energy,

$$\mathbf{F}_i = -\frac{\partial E_\theta}{\partial \mathbf{R}_i}, \qquad \boldsymbol{\sigma} = -\frac{1}{V}\frac{\partial E_\theta}{\partial \boldsymbol{\epsilon}},$$

under the declared stress sign and strain convention of the label source. MACE constructs symmetry-aware local atomic representations and sums atomic-energy contributions [1]. A useful training/evaluation corpus must therefore constrain both the energy surface and its derivatives throughout the intended simulation domain.

A low global force error is not sufficient. Common framework vibrations can dominate aggregate statistics while rare mobile-ion environments, strain states, migration geometries, interfaces, defects, or other declared focus physics remain poorly represented. The architecture separates broad numerical metrics, condition/group-resolved evidence, physical-observable validation, and explicit extrapolation/challenge evidence. The current P6 campaign ends at selected-only method validation and fresh final production; downstream qualification is separately activated.

Why trajectory frames need statistical roles

Molecular-dynamics frames are temporally correlated. Neighboring configurations can be near duplicates, so assigning them to nominally different roles can create leakage and overstate model quality.

For observable x_t, normalized autocorrelation at lag k is

$$\rho_x(k) = \frac{\langle (x_t - \bar{x})(x_{t+k} - \bar{x}) \rangle}{\langle (x_t - \bar{x})^2 \rangle}.$$

A truncated integrated autocorrelation time is

$$\tau_{\mathrm{int},x} = \Delta t \left[\frac{1}{2} + \sum_{k=1}^{k^\star} \rho_x(k) \right],$$

with approximate effective sample count

$$N_{\mathrm{eff},x} \approx \frac{T}{2\tau_{\mathrm{int},x}}.$$

mdstats therefore uses autocorrelation-aware complete-frame blocks, purge semantics, and explicit independence grades rather than treating every frame as independent [3-5]. Exact estimators, truncation, block size, purge, and role-assignment rules are specification-owned.

Evidence-role model

The architecture distinguishes evidence by what it is allowed to control.

Role	Supplies gradients?	May control fitted preparation/subset/size/checkpoint?	Purpose
development / training domain	Yes when selected	Yes, within the authorized training/model-selection contract	fitting and protocol development
checkpoint / common target monitor	No	Yes, only for explicitly authorized development/model-control decisions	stopping/checkpoint and target-size development evidence
held-out CV evaluation	No	No for the frozen protocol it evaluates	protocol validation
calibration	No	No training/subset/checkpoint changes	final-committee uncertainty calibration
locked interpolation/challenge test	No	No	sealed final evaluation

Calibration is not test data; held-out CV is not a checkpoint monitor; and a monitor cardinality is not a target-training cardinality.

Scope and ownership

The MLFF subsystem owns dataset certification, evidence-role construction, fitted preparation, one global target-size study, training-artifact construction, current campaign orchestration, checkpoint/evaluation lineage, and active-learning lineage. Downstream deployment, physical, calibration, and locked-test consumers retain their product obligations without becoming P6 selection owners.

Its current responsibilities include:

- VASP source discovery/certification and source/label identities;
- composition, thermodynamic condition, ensemble, reference-cell, strain/stress reconstruction;
- electronic-structure compatibility and label-domain grouping;
- energy/force/stress audit and atomic-reference identifiability/fitting lineage;
- immutable frame facts, eligibility, and quality decisions;
- generic raw structural features/events plus explicit optional material/profile extensions;
- autocorrelation-aware complete-frame blocks and role feasibility;
- fixed outer roles and independent CV job families;
- neutral and authorized fold-local fitted descriptors, transforms, metrics, E0, objective/weight, and difficulty evidence;
- the target-size development split, the canonical training/evaluation orders, the common preparation, and the paired optimizer-seed screen;
- one protocol-global target-size decision with one exact global selected membership;
- MACE target/replay artifacts and explicit exposure realization;
- replay-retention and checkpoint admissibility;
- post-selection protocol-matched CV and fresh final training; downstream committee, calibration, sealed evaluation, and deployment verification are separate consumer boundaries;
- active-learning candidate/DFT lineage where supported by current specifications.

The subsystem does not silently merge incompatible electronic-structure levels, infer ambiguous scientific references, use held-out/locked evidence for forbidden model-control decisions, redefine analysis-owned physical-observable algorithms, create a second target selector, generate rescue target sizes, or migrate unsupported old campaign generations.

LTA/zeolite ring, cage, site, crossing, and related semantics are optional profile extensions rather than generic defaults.

Reference application: Li/Na/K-LTA

The principal reference application contains AIMD evidence spanning multiple cation compositions, temperatures, and strain conditions. It motivates—but does not hard-code into generic architecture—several requirements:

1. framework atoms can outnumber mobile cations, so aggregate metrics must not hide declared mobile-species environments;
2. strain conditions need not form a full Cartesian product with composition/temperature, so condition applicability may be hierarchical;
3. one trajectory per condition supplies limited independence and must not be represented as an independent-replica test;
4. fixed framework stoichiometry can make individual atomic reference-energy corrections non-identifiable without anchors;
5. short trajectories may contain few rare transitions, so absent events are explicit coverage gaps rather than evidence of irrelevance.

Reuse of analysis and sampling capabilities

The MLFF workflow orchestrates existing mdstats capabilities instead of duplicating them.

Capability	MLFF use
`mdstats.io.vasp.read_vasp_frames`	cells, coordinates, energies, forces, stress, temperature, provenance
VASP control/ensemble readers	controls, energy-channel and ensemble evidence
trajectory-quality / production-regime assessment	source and stationary-regime evidence
analysis structural/topology modules	optional profile-owned raw evidence or post-training observables under analysis contracts
sampling/cross-fit primitives	source-bound blocks, purge, and independence semantics

Physical observables remain owned by `mdstats.analysis`. The MLFF layer may orchestrate matched evaluation and retain analysis-owned result identities, but it does not redefine RDF, MSD, VACF, VDOS, diffusion, topology, conductivity, or related numerical algorithms.

Current controlling data flow

```
source bytes / controls / trajectory collections
  -> source and label-domain certification
  -> immutable frame facts and eligibility
  -> raw features/events before ordinary thinning
  -> correlation-aware blocks and evidence-role feasibility
  -> development / monitor / post-selection CV roles
  -> neutral DATA6/DATA7 fitted preparation
  -> P_train / M3 split -> pi_train / pi_eval
```

```
  -> common target-size preparation
  -> paired optimizer-seed screen -> target-size reducer
  -> target-size study using authorized development/model-selection evidence
  -> one frozen N_selected and exact global T_selected
  -> protocol-matched CV partitions inside T_selected, with held-out folds inaccessible to
size/checkpoint choice
  -> accepted frozen protocol
  -> independent final seeds and checkpoint admission
  -> current final-production publication
  -> separately activated downstream committee/physical/calibration/locked consumers where
implemented
  -> active-learning lineage where configured
```

No allowed dependency runs from held-out CV or locked-test evidence backward into fitted transforms, E0 fitting, target membership, target-size selection, checkpoint choice, or calibration-policy design.

Responsibility separation is more durable than module layout

The implementation may reorganize Python modules while preserving the architecture. The durable separation is among:

- physical/source facts;
- evidence-role and policy decisions;
- training-domain fitted products;
- target-membership and target-size decisions;
- runtime/execution realization;
- validation/calibration/locked evidence;
- external analysis-owned results.

Current specifications control public/serialized current-generation contracts. Internal refactoring may reuse common sampling/execution primitives when externally owned scientific behavior and persisted current-generation identities remain conforming. Backward compatibility with superseded campaign generations is not an architectural requirement. The accepted current-generation P5A6 workspace remains a required unchanged reopen boundary; obsolete derived target-size generations are rejected before reuse and current preparation creates a fresh configurable authority.

Part II - Data and evidence contracts

Purpose and ownership

This chapter defines immutable source/frame facts, label-domain identity, physical conditions, quality/eligibility, raw feature/event providers, and correlation-aware complete-frame evidence blocks.

It does not own evidence-role assignment beyond the records needed to support DATA5, fitted statistics, target membership, target size, training exposure, checkpoint selection, or validation decisions.

Evidence records and immutability

The MLFF data model separates source facts, workflow decisions, fitted products, runtime realizations, and external scientific results. A new policy creates new policy/decision records rather than mutating immutable source/frame facts.

Source and frame facts

`TrainingDataSource` owns source occurrence identity, path/location hints, content hashes, composition/controls, ensemble/quality/production evidence, label-domain identity, and declared reference grouping.

`TrainingFrameRecord` owns source-bound frame facts such as `frame_uid`, source occurrence, frame index/time, atoms/cell, label references, physical conditions, and distinct geometry/label fingerprints.

`TrainingFrameRecord` does **not** own eligibility, statistical role, target membership, target size, training exposure, calibration, or acquisition state.

Decision, policy, fitted, and realization families
Representative downstream/current families include:

```
FrameEligibilityDecision
PartitionAssignment
CandidateAdmissibilityDecision
AcquisitionDecision

PartitionRoleBudgetPolicy
PartitionFeasibilityReport
FeatureMetricPolicyTemplate
FoldFeatureMetricFit
FinalFeatureMetricFit
TrainingObjectivePolicy
ConfigurationWeightPolicy
PropertyWeightPolicy
CheckpointMetricPolicy
TargetSizeExperimentDefinition
ResolvedTargetSizePolicy
TargetSizeCommonPreparation
TrainingProtocolIdentity
MaceCheckpointControlPolicy
ExposureBackendPolicy
MaceExposureRealizationRecord
ReplayRetentionPolicy
ProtocolFreezeRecord
CalibrationApplicabilityDomain
CalibrationTransferDecision
```

A static policy defines an algorithm and fixed choices. A fitted record contains parameters learned from one explicitly authorized training domain. A realization record records behavior actually observed from an external/runtime system. Those roles are not interchangeable.

Target membership is intentionally absent from the generic DATA2-DATA4 record list because its current authority is the canonical training order `pi_train` derived by the target-size owners in Part V.

Digests and signatures
Deterministic content/policy/source digests bind identity and detect modification. They do not by themselves authenticate authorship. Serialized current records carry version/schema and deterministic content identity under their owning specifications.

Source manifest and occurrence identity

A review/production manifest supplies source locators, grouping declarations, scientific assertions that cannot be reconstructed unambiguously from one source file, and explicit expert overrides with rationale. Directory/file naming is diagnostic input, not accepted physical truth without verification.

The source byte/content identity is distinct from a manifest occurrence. Byte-identical copies may share a source-content identity while deliberately distinct manifest runs have distinct occurrence identities.

A frame occurrence derives from occurrence identity plus source frame index. This keeps occurrence identity stable across later concatenation/export while permitting duplicate-geometry detection across separate source occurrences.

Geometry, label, and labeled-configuration identities

The architecture keeps three identities separate:

```
geometry_fingerprint
label_payload_digest
labeled_configuration_fingerprint
```

`geometry_fingerprint` identifies atomic geometry independently of energy/force/stress labels under the current canonical wrapping/cell/tolerance policy. `label_payload_digest` binds the selected labeled payload and label-domain identity. `labeled_configuration_fingerprint` combines geometry and label payload.

Leakage auditing may use occurrence overlap, exact geometry overlap, exact labeled-configuration overlap, declared near-geometry/descriptor criteria, restart/copy detection, and forbidden temporal proximity. Approximate or symmetry-aware matching may exist only under an explicit current policy; it cannot change the semantic roles above.

Electronic-structure label domains

Electronic-structure identity is decomposed because not every input difference has the same scientific meaning:

```
TheoryIdentity
EnergyReferenceIdentity
DerivativeConvention
NumericalQualityProfile
SoftwareProvenance
```

A versioned `LabelCompatibilityPolicy` classifies differences as compatible, compatible with quality evidence, separate label domain, or unresolved. Theory- or reference-defining differences cannot be silently waived.

A target training bundle contains one compatible target label domain and, when enabled, a separately identified replay head/lineage. Incompatible target DFT levels produce separate target bundles rather than an implicit mixed target domain.

Energy channel

The selected target energy is an explicit named channel consistent with derivative labels. Its channel, units, completeness, electronic/reference convention, and provenance are preserved. Energy/force/stress labels that do not share an accepted derivative/reference convention are not silently combined.

Atomic-reference identifiability and fitting boundary

For elemental correction vector $\Delta\mathbf{e}_0$, the schematic fit is

$$\mathbf{A}\,\Delta\mathbf{e}_0 \approx \mathbf{b},$$

with configuration-element count matrix $\mathbf{A}$ and target-minus-foundation energy residual $\mathbf{b}$.

`AtomicReferenceIdentifiabilityReport` depends on elemental count support rather than fitted target residuals. It records element order, matrix shape/rank/singular values, condition/null-space information, identifiable combinations, outcome, and transfer limitations. It does not contain fitted elemental corrections.

The actual `AtomicReferenceFitRecord` is a DATA7 fitted object bound to one fold/final training domain, foundation checkpoint identity, identifiability report, solver/tolerance, elemental support, fitted corrections, residual, and policy outcome.

Each cross-validation fold receives its own fold-local fit. Final training receives a separate final-training fit. Monitors, calibration, held-out evaluation folds, and locked tests are excluded from the fit.

MACE export receives the exact accepted numerical E0 representation, normally an atomic-number mapping; a record/path name is provenance rather than an E0 payload.

Ensemble, temperature, cell, and strain

Ensemble and temperature

The subsystem consumes mdstats control/ensemble certification and distinguishes equilibrium, pressure-controlled, ramped/driven, multi-thermostat, and unresolved cases under the owning control specification. Ensemble is not inferred merely from observed cell variation.

Nominal temperature controls and realized ionic-temperature statistics remain separate. `TemperatureCondition` binds requested/thermostat targets, realized series/statistics, drift/stationarity evidence, and ramp status as applicable.

Reference-cell resolution

Strain requires an explicit or uniquely resolvable compatible reference. Accepted resolution order is controlled by current specifications and may use an explicit matrix/structure/run or a unique compatible unstrained member of the declared reference group. Ambiguity fails closed.

Cell and deformation convention

ASE cell vectors are rows. For fractional row vector $\mathbf{s}_{\text{row}}$ and cell $\mathbf{H}$,

$$\mathbf{r}_{\text{row}} = \mathbf{s}_{\text{row}}\mathbf{H}.$$

For reference $\mathbf{H}_0$ and current cell $\mathbf{H}_t$, the reported deformation gradient acting on Cartesian column vectors is

$$\mathbf{F}_t = \left(\mathbf{H}_0^{-1}\mathbf{H}_t\right)^T.$$

An internal right-acting row-vector form is acceptable only when serialization/reporting returns the declared Cartesian-column convention. Rotation/stretch separation uses the declared polar-decomposition convention.

Stored strain evidence includes the applicable volume, linear/finite/logarithmic, hydrostatic/deviatoric, principal, shear, rotation, coordinate-frame, and storage-convention quantities. Qualification includes nonsymmetric shear and rotated-stretch cases so transpose/left-right errors cannot hide behind diagonal fixtures.

Hierarchical condition schemas

Condition space is not assumed to be a global Cartesian product. A material profile declares applicable condition axes and hierarchical strata. For example, an LTA profile may separate unstrained composition/temperature/regime strata from strained composition/reference-condition/strain-mode/sign/regime strata. Only observed and scientifically applicable combinations are required.

Stress and virial

Canonical `REF_stress` is a symmetric Cartesian Cauchy-stress tensor in eV/Angstrom^3 under the ASE/MACE sign convention qualified by the runtime lock. Tensor shear carries no engineering-factor multiplication. Intermediate Voigt ordering, when used, is explicit and round-tripped.

Virial and stress have distinct keys and are never silently relabeled. Qualification covers units, finite-strain sign, tensor/Voigt order, shear factors, and MACE read-back. Missing stress may carry zero stress weight only under an explicit heterogeneous-label policy.

Eligibility and quality

Run/source quality distinguishes qualified, degraded, unqualified, and unresolved states under current policy. Overrides are explicit evidence.

`FrameEligibilityDecision` applies after labels exist. Hard rejection includes absent/nonfinite required labels or geometry/cell, singular/corrupt structures, incomplete ionic records not recoverable under the current interruption policy, catastrophic overlaps, and disallowed electronic-convergence failures.

Soft evidence records transient regimes, unusual but physical forces/stress, rare coordination/events, topology changes, model residuals, and degraded numerical quality without turning percentile tails into automatic rejection.

Pre-DFT candidates use a separate `CandidateAdmissibilityDecision` over geometry/cell safety, element/count policy, topology/integrity, trajectory/integrator evidence, model outputs, and descriptor availability. After DFT labeling they re-enter normal source/frame eligibility lineage.

Material profiles and feature providers

Declarative profile boundary

`SystemProfileProvider` owns material identity: phases, geometry, chemistry modifiers, optional structural extensions, meaningful atom groups, condition axes, and independence axes. It does not itself own calculated scientific feature arrays.

Profiles are compositional rather than a single flat material enum. Interface/multiphase systems explicitly declare component membership. Generic fallback supplies only generic groups/axes; porous/zeolite/LTA semantics require the corresponding explicit extension chain and never activate automatically.

Universal structural selection inputs

The generic structural provider supplies selection-grade local geometry descriptors such as smooth chemistry-scaled coordination, support-neighbor count, radial projections, local-density/mixing proxies, angular moments, and rotationally invariant orientational-order summaries.

These are upstream target-subset inputs, not an independent selector and not replacements for analysis-owned RDF, integer coordination, full angle distributions, structure factors, or topology observables.

Descriptors aggregate by authorized atom groups and elements present in the permitted domain. Generic temporal events capture large local structural changes without assigning material-specific physical meaning.

Partition-critical profile features

Rare categorical states that partition policy promises to protect are available at full resolution before the outer partition freezes. A profile may supply phase, environment, defect, region, molecular, or event states.

Optional LTA state includes framework/mobile roles, resolvable ring/site class, off-center class, coordination/site/ring-crossing changes, and framework-integrity evidence. Unresolved required classifications produce explicit coverage/partition limitations rather than fabricated balanced strata.

Optional learned-model features

A qualified optional MACE provider may supply foundation/model identity, invariant atomic descriptors, group/species environment summaries, and authorized zero-shot prediction/residual evidence. MACE/PyTorch remain optional dependencies to the mdstats core.

Feature blinding and fitted metrics

Geometry-only descriptors from a frozen model may be computed wherever authorized. Label-derived residual/difficulty features may be exposed only inside their applicable training domain.

Outer monitor, calibration, held-out evaluation, and locked-test residuals do not enter feature fitting or target-subset construction. Evaluation predictions can be persisted in blinded catalogs without exposing residual-derived selector inputs. Violating this boundary is a hard leakage failure.

Raw feature providers are partition-independent. Dataset-dependent scaling/PCA/whitening/metric fitting is represented by separate static templates and fold/final fitted records. For fold k, fitting may inspect only its gradient-training domain; other domains may be transformed by the frozen fitted object but cannot influence it.

A block-normalized metric may take the form

$$d^2(i,j) = \sum_b w_b \frac{\| \mathbf{z}_i^{(b)} - \mathbf{z}_j^{(b)} \|_2^2}{d_b},$$

where block weights, dimensions, missing-block behavior, dtype, tolerance, and any fitted scaling/projection are explicitly identified. High-dimensional learned descriptors do not dominate solely because they contain more components.

Event detection before thinning

The controlling order is:

1. source/label integrity on full-resolution frames;
2. event/change detection on all eligible frames;
3. protected event-window preservation;
4. temporal thinning of the ordinary non-event pool;
5. higher-cost descriptor/fitted target-subset-input operations.

Event stencils are policy-controlled and compact by default. Adjacent frames from one physical event are not mistaken for independent rare-event evidence.

Autocorrelation and complete-frame blocks

Fast observables determine the minimum ordinary decorrelation/block scale; slow structural variables diagnose whether state-level independence exists at all. Candidate stride is defined in physical/frame time from the declared fast autocorrelation estimate and applies only to the non-event pool.

A `TrainingDataBlock` is a contiguous interval of whole configurations with block/run/frame bounds, represented time, regime, correlation evidence, and configuration identities. Atoms from one configuration are never split across statistical roles.

A purge interval separates roles using the declared physical/autocorrelation/event/restart policy. If a slow state never decorrelates, the independence report explicitly states that temporal blocking does not provide state-level independence.

Part III - Statistical design and fitted preparation

Purpose and ownership

This chapter defines the evidence roles and fitted-preparation boundary that make target-size comparisons and later method validation interpretable. It owns independence, protected relations, leakage boundaries, fitted products, objective/weighting inputs, and the distinction between development evidence and later validation roles.

It does **not** own target membership or target size. The Part V owners derive one `P_train`/`M3` split, one canonical `pi_train`, and one target-size result. After selection, the Part V/P5 owners may partition the already frozen `T_selected` for cross-validation; that operation cannot choose a new size or membership.

Independence and evidence roles

Evidence uses the strongest available independence level, for example:

1. independent replica/velocity seed or independently prepared realization;
2. independent structural or chemical ordering;
3. independent thermodynamic run;
4. a purged temporal block within one run.

Temporal separation does not create an independent metastable state when the relevant slow variable has not decorrelated. Every cohort carries machine- readable independence evidence and known limitations.

Before roles are assigned, the partition policy declares requested cohorts, minimum independent blocks, purge requirements, protected relations, and allowed reductions. A feasibility report may therefore record full support, temporal-block-only support, deferred calibration, external-only challenge evidence, a reduced fold count, or insufficient support. The workflow never fabricates a role from a short or correlated trajectory to satisfy a percentage target.

The current development evidence roles are:

```
development_pool
common_target_monitor
post_selection_cv_folds
```

The broader product architecture reserves separate calibration and locked-test roles for downstream qualification. Those consumers are not part of the P6 campaign lifecycle and their absence is not converted into current selection or production evidence.

Only the development pool supplies gradient-training candidates. The common target monitor is development/model-selection evidence: it may control the authorized target-size screen and post-selection checkpoint policy, but it supplies no gradients and is not a held-out CV fold.

One global selection universe

The neutral statistical substrate supplies duplicate groups, correlation families, provenance relations, and split exclusions before target-size selection exists. It produces exactly one development split:

```
eligible labelled frames -> P_train (target-training pool) + M3 (development monitor)
```

`P_train` is ordered once as `pi_train`; the target-size owner defines every candidate as an exact prefix. `M3` is ordered once as `pi_eval`; `M1`, `M2`, and `M3` are direct nested evaluation populations. No complement, per-domain membership map, or alternate ordering may change the universe.

Protected relations remain intact wherever the current owner assigns roles. An inseparable duplicate/correlation component cannot be split merely to obtain a requested fold count. A frame outside `T_selected` cannot enter post-selection CV because it is convenient or because it belongs to a related source cohort.

Cross-validation validates a frozen protocol

Target size is frozen before protocol-matched cross-validation is interpreted. For each required post-selection fold (k), the owner keeps distinct:

```
fold_training_partition_k within T_selected
fold_checkpoint_monitor_k
held_out_evaluation_partition_k within T_selected
```

The selected cardinality and the exact global membership remain unchanged for every fold. Fold assignment may be local to `T_selected`, and fold-local fitted preparation may use only that fold's training partition and authorized monitor. It may not inspect the held-out partition, outer protected evidence, or locked evidence before checkpoint choice. The final fold evaluation occurs only after the fold representative is frozen.

This gives the required distinction:

```
global target-size choice -> one N_selected and one T_selected
post-selection CV        -> method validation on partitions of T_selected
```

Held-out CV error, calibration evidence, and locked-test evidence therefore cannot select `N_selected`, alter `T_selected`, or tune the target-size policy.

Fitted preparation

The current common preparation is built once from the neutral substrate and the frozen foundation/training protocol. It may emit:

- descriptor coordinates and fitted feature metrics;
- foundation predictions and training-domain residual/difficulty evidence;
- atomic-reference/E0 fits;
- objective, configuration-weight, and property-weight records;
- condition, provenance, event, environment, and diversity inputs;
- deterministic identities binding each product to its authorized inputs.

These products are inputs to the one canonical training order. They are not a second selector. A fitted transform, metric, residual, or E0 correction must be bound to the evidence that fitted it and may not be inferred from a downstream held-out result.

For post-selection CV, a fold-local transform or metric is valid only when the CV owner explicitly records the fold training partition, protected relations, and protocol identity. A fold-local product can change the fold's evaluation realization; it cannot change the global target membership or target-size decision. Final production uses the accepted method and complete `T_selected`.

Selection inputs are not a second selector

Representative density, diversity, environment coverage, protected events, difficulty, condition balance, and provenance/correlation structure remain useful scientific information. The current owner represents them as:

```
fitted feature coordinates/metrics
hard obligations or applicability masks
representative-density and diversity evidence
event/environment/condition evidence
difficulty and correlation identities
```

The target-size policy combines these inputs into the one deterministic `pi_train`. There is no competing quota/FPS plan whose prefixes can disagree with that order. A materialization or export record may describe a consumer view of `T_selected`, but it is not an independent membership authority.

Objective, weighting, and exposure

Target membership, target size, loss weighting, and runtime exposure are separate decisions. `TrainingObjectivePolicy` binds the loss family, energy/force/stress weights, head weights, normalization, robust-loss choices, and missing-label behavior. Configuration/property weighting binds applicable condition, regime, event, quality, and property weights. Exposure binds the head, actual gradient exposures, batching/duplication behavior, seed, and runtime lineage.

A frame can be selected once, weighted non-uniformly, and exposed through a qualified loader without those decisions becoming one authority. A custom atomwise or auxiliary loss changes `TrainingProtocolIdentity` and requires its own accepted method identity; it cannot be smuggled into the current protocol through a loader option.

Material and profile specialization

Condition axes and focus groups are declared by the applicable material/profile contract. They may include composition, temperature, pressure, strain, phase, defect, surface/interface state, conformer, preparation history, or another scientifically justified axis. A profile may define hierarchical applicability rather than a Cartesian product. Empty or physically inapplicable combinations are not missing observations merely because their names exist.

Material-specific concepts remain explicit extensions. LTA ring/cage/site groups or Li/Na/K focus groups are not generic defaults and cannot silently change the global target order.

Dependency boundary and failure semantics

The allowed dependency direction is:

```
raw source / label / feature / event evidence
    -> neutral statistical substrate and protected relations
    -> P_train/M3 split and canonical orders
    -> common fitted preparation
    -> one target-size screen and reducer
    -> frozen N_selected/T_selected
    -> post-selection fold partitions and method acceptance
    -> fresh final production
    -> downstream qualification roles when separately implemented and activated
```

Forbidden reverse dependencies include held-out CV error choosing target size, locked evidence tuning preparation or checkpoint policy, calibration fitting the protocol it evaluates, and executor/cache behavior changing membership or evidence roles.

The workflow fails closed when labels or protected relations are unresolved, requested roles are infeasible, a fitted product has the wrong lineage, a fold would split an inseparable relation, or a downstream result is offered as selection authority. Explicit absence or deferral is evidence; it is not a synthetic pass.

Part IV - Training, evaluation, and downstream qualification boundary

Purpose and ownership

This chapter defines the training-protocol identity, replay boundary, checkpoint admissibility, post-selection cross-validation, and fresh final production consumed by the current campaign. Target membership and target size are already frozen by Part V; this chapter never creates a second size or membership authority.

Deployment, physical-observable comparison, uncertainty calibration, and locked testing remain product capabilities, but their downstream qualification consumers are outside the P6 public lifecycle. They may consume a current final-production publication only through a separately implemented and explicitly activated successor contract. They cannot feed selection or choose another final model.

Complete training-protocol identity

Multi-head replay fine-tuning trains a shared MACE backbone on target data and an authorized foundation replay corpus with separate output heads. Replay can constrain forgetting while the target head adapts, but replay evidence is not a target-size ranking signal.

Every compared run binds a complete `TrainingProtocolIdentity`, including as applicable:

```
foundation checkpoint / model family / selected foundation head
protocol-global N_selected and exact T_selected binding
replay source, split, and replay-monitor identity
training objective and configuration/property weights
target/replay head weights and realized exposure policy
checkpoint metric and admissibility policy
optimizer, LR schedule, epoch cap, stopping policy, and seed policy
model precision, acceleration backend, and MACE adapter/runtime lock
```

The identity contains no unbound caller-held model or fold result. A change to replay semantics, objective, selected membership, checkpoint policy, precision/backend, stopping/LR policy, or another protocol field creates a new method identity and invalidates the descendants that depend on it.

Target and replay evidence

Target and replay retain separate source/label identities, split and exposure accounting, weights, and monitors. Replay preparation never silently acquires an external corpus. True-label replay is compared against its authorized labels; pseudo-label replay, when explicitly supported by the method contract, measures drift from the bound foundation model on an unseen monitor.

`ReplayRetentionPolicy` binds its metric, baseline, permitted degradation, aggregation, and failure semantics. A checkpoint that violates a mandatory replay-retention requirement is inadmissible even when its target metric improves. Replay values receive no target-size ranking, tie-break, fold, or seed credit.

Monitoring and checkpoint choice

The common target monitor is development/model-selection evidence. It supplies no gradients and is distinct from post-selection held-out fold evidence and from future locked-test evidence. Monitor cardinality is never target-size authority.

`CheckpointMetricPolicy` defines the primary target objective and every mandatory target, focus-group/species, condition, property, replay, and integrity constraint applicable to checkpoint admission. A typical constrained choice is

$$\min_c L_{\text{target monitor}}(c)$$

subject to requirements such as

$$L_{F,g}(c) \leq \delta_g, \qquad \Delta L_{\text{replay}}(c) \leq \delta_{\text{replay}}.$$

Exact thresholds and aggregation are specification-owned serialized policy. Checkpoint choice is deterministic over the complete authorized candidate set and fails closed when no candidate satisfies a mandatory constraint.

MACE adapter and data boundary

The MACE adapter binds package/source identity, head ordering, loader realization, scheduler/stopping behavior, checkpoint retention, precision/backend realization, and any current runtime lock. Documentation URLs are not a runtime contract. Material upstream behavior changes fail closed until the adapter contract is revised and requalified.

Extended XYZ contains only MACE-readable labels, weights, and compact stable identities. Sidecar manifests carry long provenance, policy identities, and audit reasons. Target export includes the declared energy channel, forces, authorized stress, configuration/property weights, cell/PBC, atom order, and exact label/E0 provenance. Export precision and round-trip behavior are checked through the current reader path.

An `AtomicReferenceFitRecord` becomes the explicit numerical representation accepted by the MACE runtime, normally an atomic-number mapping. A record name or path is not an E0 payload. Target and replay label domains are checked for compatibility rather than silently merged.

Controlled target-size screen versus ordinary training

The target-size experiment is the special Part V protocol-comparison control. It uses authenticated `n1 -> n2 -> n3` continuation, paired optimizer seeds, direct `M1/M2/M3` endpoint populations, and no ordinary target-success early stopping before a required screen boundary. An earlier checkpoint cannot replace the prescribed endpoint merely because its metric is better.

The current public screen owns the complete restartable continuation. Generated campaigns default to `(n1,n2,n3) = (1,3,10)`; fresh final production has its independent `[training].max_num_epochs` horizon. Screen checkpoints and CV checkpoints are never production parents.

After selection, CV and final production run under the accepted method. CV uses fold partitions of exactly `T_selected`, with fresh model/optimizer lineage per required fold/seed. Final production starts fresh from the accepted foundation and trains the complete `T_selected`; it continues no screen or fold trajectory. Its run namespace remains disjoint even when a numeric seed or target size coincides.

Post-selection method acceptance

The dependency graph is acyclic:

```
current selected binding
  -> shared post-selection method identity
  -> CV policy and final-production policy
  -> CV plan and final-production plan
  -> fold/final execution and evidence
```

The shared method identity binds preparation/objective recipe, foundation and initialization family, optimizer family, LR schedule, checkpoint semantics, precision, and backend. It does not contain fold membership or a second target size.

The CV policy owns `K >= 2`, partition seed, fold algorithm, CV budget, monitor/purge allocation, target-only acceptance, and the all-required-fold / all-required-seed rule. The final-production policy owns the production epoch horizon, production seed matrix, and committee policy. Neither policy can rewrite the other or the selected binding.

The CV plan records the current selected binding, protected P1 relations, selected-only fold memberships, and required run matrix. The final-production plan records the complete `T_selected` and accepted CV authorization. Evidence descends from a plan and binds it; corrupted evidence invalidates itself and never rewrites its authorizing plan.

CV freezes each fold representative on its authorized target monitor before evaluating the held-out fold. A required fold or seed failure is a methodological failure: it leaves `N_selected` and its evidence unchanged and does not authorize final production. A materially different method requires a new target-size experiment because the measured method has changed.

Final production and currentness

Final production publishes only after reauthenticating the current campaign revision, selected binding, accepted method, and complete production plan. `ProtocolFreezeRecord` binds the method, selected membership, replay/monitor identities, checkpoint/committee identities, and upstream evidence needed by the current production consumer.

Every current read resolves the selected binding again from the store; it does not trust a stale caller object. Publication rechecks currentness in the same transaction that would make a descendant current. A superseded run can retain diagnostics, but it cannot publish a current final model.

Downstream product boundary

Physical observables such as RDF, coordination, topology, MSD, VACF, spectra, VDOS, diffusion, and conductivity remain owned by their analysis modules and their own specifications. A future downstream qualification recipe must bind matched reference/candidate collection identity, runtime/capability identity, analysis-owned result identity, and an explicit statistical role.

Calibration is valid only for predictions from the actual frozen final committee. Locked-test evidence remains sealed until its explicit activation boundary. Neither calibration nor locked evidence may alter fitting, target membership, target size, training protocol, checkpoint selection, or final publication. P6 does not claim that these downstream consumers are implemented or qualified.

Failure and reproducibility semantics

The workflow fails closed for incompatible label domains, missing foundation or replay identity, unsupported loader exposure, missing required fold/seed, stale selected binding, invalid checkpoint constraints, corrupt checkpoint state, or a downstream result offered as selection authority.

Reproducibility binds source/label and protected-role identities, the neutral substrate, target-size experiment and orders, common preparation, selected binding, method/policy/plan identities, replay/monitor identities, optimizer/LR/stopping/seed policy, precision/backend, checkpoint evidence, and published final identity. Worker count, queue order, cache path, and other execution-only choices remain outside scientific identity unless a current specification explicitly says otherwise.

Part V - Target-size selection and post-selection validation

Purpose and ownership

This chapter defines how the campaign decides **how much labelled data a training method needs**, and how that decision is validated afterwards without contaminating it.

The current chain is:

```
canonical frame authority (Part II)
    -> neutral statistical substrate (Part III)
    -> one P_train / M3 target-size development split
    -> one canonical training order pi_train
    -> one canonical evaluation order pi_eval with nested M1 subset M2 subset M3
    -> one common deterministic target-size preparation
    -> paired optimizer-seed screen over candidate sizes
    -> one target-size reducer
    -> N_selected and T_selected = pi_train[:N_selected]
    -> post-selection cross-validation on exactly T_selected
    -> fresh final production on the complete T_selected
    -> currentness-fenced publication
```

Each element has exactly one owner. The reducer is the only authority that may declare a selected size; `CampaignStore` is the only authority that holds the current selected set; post-selection cross-validation is the only authority that accepts or rejects the training *method*; and final production is the only authority that publishes a production model.

There is no alternate selection path. The retired per-domain multi-view chain (compatibility-domain role freezes, full-pool feasibility, exact sparse neighborhood indices, progressive multi-view ordering, repaired master orders, continuation-state families, and independent prefix qualification) is not a current architecture, is not migrated, and is not reachable from any current runtime owner. Workspaces holding that derived state are rejected with an actionable destructive-reset requirement rather than translated; see Part VII.

Why target size is decided by a screen, not by coverage

The scientific question is empirical: *at what training-set cardinality does the accepted training method stop improving materially on a representative held-in evaluation population?* That is a property of the method, the data distribution, and the optimizer - not of a geometric covering argument over descriptor neighborhoods.

Four principles control the design:

1. one deterministic training order, so candidate sizes are exact nested prefixes and size comparisons are never confounded by resampling;
2. one common preparation shared by every candidate size and optimizer seed, so preparation cannot become a hidden per-size variable;
3. only the ordered optimizer-seed set is the stochastic replicate dimension of the screen;
4. the decision consumes target-side metric evidence alone, and downstream replay, cross-validation, physical, or deployment evidence can never rank or tie-break a size.

The development split and the two canonical orders

The neutral statistical substrate supplies protected relations - duplicate groups, correlation families, and split exclusions - before any target-size object exists. From it the architecture derives exactly one split:

```
eligible labelled frames -> P_train (target-training pool) + M3 (evaluation pool)
```

`P_train` is ordered once into `pi_train`. A candidate of nominal size `N` is the exact prefix

$$T_N = \pi_{\text{train}}[: N].$$

`M3` is ordered once into `pi_eval`, and the evaluation ladder is the nested family

$$M_1 \subset M_2 \subset M_3,$$

taken as direct prefixes of `pi_eval`. Rungs are direct populations, never complements of one another: a rung is evaluated on exactly the frames it names.

Both orders are deterministic functions of the canonical frame authority, the neutral substrate, and the configured target-size policy. Neither depends on any compatibility grouping, label-domain identity, or cross-validation plan.

The common preparation

One `TargetSizeCommonPreparation` identity is frozen before any candidate trajectory starts. It derives from `P_train` and the configured foundation/training protocol, and it is shared unchanged by every candidate size and every optimizer seed.

It MUST NOT derive from `M1`, `M2`, `M3`, held-out evidence, calibration evidence, locked evidence, or any cross-validation plan. A change to the common preparation is a change to the target-size scientific identity and produces a new generation rather than an in-place edit.

The paired optimizer-seed screen

For every candidate size `N`, the screen runs the same ordered optimizer-seed set - by current policy the two seeds `[1, 2]` - through the same fidelity ladder:

```
n1 / M1  ->  n2 / M2  ->  n3 / M3
```

Fidelity boundaries are continuation points, not restarts: model, optimizer, and RNG state continue exactly across `n1 -> n2 -> n3`. Ordinary early stopping may not truncate a required screen boundary, and the seed set is identical at every `N` so a size comparison is never a seed comparison.

Candidate rungs execute through the accepted TRAIN2 runtime and are evaluated through the accepted EVAL2 owners. Expensive numerical training has exactly one substitution seam, strictly below the mdstats owner boundary; configuration resolution, authority construction, materialization, provider and checkpoint authentication, publication, reconciliation, and adoption are production code in every invocation.

The reducer and the terminal decision

One reducer consumes the screen evidence and advances the experiment. Its outcome is one of:

- **selected** - a size `N_selected` is frozen together with the exact membership `T_selected = pi_train[:N_selected]`;
- **typed scientific terminal failure** - the configured candidate ceiling did not converge, too few candidates qualified, or the surviving candidates were not comparable.

A configured-ceiling nonconvergence is a typed result, not an invitation to invent a rescue size outside the configured ladder.

Ranking is owned by the target-side metric and practical-equivalence policy alone. Inside the practical-equivalence band the **smaller** `N` is preferred, because the scientific question is the smallest sufficient training-set size.

Currentness and the selected set

`CampaignStore` holds one canonical target-size generation. Its durable regimes are `legacy`, `transitioning`, and `current`; only `current` executes target-size work. Every mutation is one compare-and-set transition against the exact predecessor revision, so an interrupted operation is owned by the persisted transition rather than by the process that began it.

The terminal projection binds `N_selected` and the exact `T_selected` membership digest together; neither may be edited independently, and a reload re-derives the projection from the authenticated reducer state and training order rather than trusting the stored copy. Terminal currentness is always established from the current store revision, never from a caller-supplied snapshot, and a public terminal view is re-authenticated at exposure time so a stale object cannot be published after the store advances.

Invalidation scope

A change to target-size scientific identity - source or frame membership, canonical numerical labels or their interpretation policy, the candidate ladder or configured ceiling, the evaluation-size ladder, fidelity boundaries, the ordered optimizer-seed set, the training-order policy, the `P_train/M3` split or `pi_eval` ordering policy, the common preparation, the metric/practical-equivalence policy, or the foundation/replay identity where it is part of the experiment - replaces the generation. The old selected set stays readable as history and can never re-enter current authority.

Changes that are *not* target-size identity invalidate only their own descendants:

- advisory provenance grouping or report presentation invalidates only the advisory evidence that depends on it, and never the frame UID, the canonical label identity, the neutral partition, or the target-size result;

- cross-validation-only settings such as fold count and partition seed invalidate cross-validation and its descendants, and leave `N_selected`/`T_selected` byte-identical;
- production-only budget or runtime policy invalidates only final-production descendants.

Post-selection cross-validation

Cross-validation starts only after the terminal selection is frozen, and it consumes exactly `T_selected` - complete coverage, no unselected sibling frame, no held-out outer frame.

It validates the **training method**, not the size:

- the configured fold count `K >= 2` and every required fold of every required CV seed must pass the configured target-only acceptance predicate; there is no mean, majority, best-seed, partial, `K = 0` or `K = 1` authorization;
- the full P1 split-exclusion and correlation-family constraints continue to hold inside fold assignment;
- fold-local preparation, training, checkpoint selection, and replay admissibility may never see that fold's held-out outer target set, and the fold representative freezes before held-out outer evaluation;
- replay training exposure and the TRUE_DFT replay admissibility monitor remain distinct concerns, and TRUE_DFT replay contributes no ranking, tie-break, fold, or seed credit;
- a cross-validation failure is a methodological result: `N_selected` and its evidence are unchanged, and final production is simply not authorized. If cross-validation shows that a materially different training method is required, that changed method needs a **new** target-size experiment, because the method whose convergence was measured has changed.

Supported training modes remain exactly `scratch`, `naive_fine_tuning`, and `multihead_replay`; the canonical post-selection heads remain `target_head` and `pt_head`; and the foundation checkpoint head remains a separate foundation-owned concept. Method, foundation, replay, and content identity all fail closed.

Fresh final production

Final production starts fresh from the accepted foundation/initialization with fresh optimizer, RNG, and run state. It trains on the complete exact `T_selected`, under the cross-validation-accepted method, for the configured `[training].max_num_epochs` - an independent production horizon that is deliberately unrelated to the screen's `n3`.

Frozen `M3` evidence may remain development/model-selection evidence. Final authorization and publication remain currentness-fenced and restart-authenticatable: a reopened campaign reauthenticates the selected binding, the cross-validation acceptance, and the final publication identity before exposing any of them as current.

Public command surface

The current lifecycle, including configuration initialization, is exactly:

```
init -> doctor -> prepare -> select-target-size -> cross-validate -> train-production
```

`prepare` reconstructs the current substrate and cannot select a size. `select-target-size` is the only command that trains candidates and decides `N`. `cross-validate` is the only command that accepts the method. `train-production` is the only command that publishes a fresh production model. `status` and `advance` project this lifecycle from the owning authorities rather than from stage markers.

Part VI - Bounded execution, restart, and performance architecture

Purpose and authority

Execution optimization is acceptable only when it preserves the scientific and statistical authorities in Parts I-V and improves measured throughput, memory, storage, or restart cost. Utilization is diagnostic; authenticated records, deterministic decisions, and exact scientific digests decide correctness.

Worker count, queue depth, query-block size, cache location, file-backing threshold, storage path, and similar execution choices do not enter scientific identity unless a current specification explicitly makes them part of the algorithm.

The central rule is:

> change how exact work is scheduled or represented, not what evidence is consumed or what authoritative decision is produced.

Work/span and single-level parallelism

For serial work (T_1), critical path (T_), and (P) admitted CPU lanes,

$$T_P \geq \max\left(\frac{T_1}{P}, T_\infty\right).$$

Independent work is exposed at the highest useful level. Nested numerical parallelism is suppressed while outer work fills the resource budget:

$$P_{\text{outer}} P_{\text{native}} \leq P_{\text{budget}}.$$

The resource scope controls cKDTree, BLAS, OpenMP, PyTorch, and other native threads. A process or worker does not independently oversubscribe the host. The implementation may use exact kernels such as `query_ball_point`, `numpy.bincount`, bounded indexed reductions, or `threadpoolctl`; these are execution realizations and do not change evidence roles or canonical order.

Preparation and execution boundaries

The source/frame and numerical-label authorities are built once and validated through their current owners. The neutral statistical substrate supplies the one `P_train/M3` split and the two canonical orders. The current P3 common preparation is then computed once and shared by all authorized candidate sizes and optimizer seeds.

```
source/frame/label authorities
  -> neutral statistical substrate and protected relations
  -> one P_train/M3 split and pi_train/pi_eval
  -> one common target-size preparation
  -> paired-seed candidate screen
  -> selected binding
  -> selected-only CV and fresh final production
```

The common preparation is a single authenticated authority, not one independent copy per candidate or fold. A post-selection CV fold may create a fold-local fitted view from its own training partition when its owner requires it, but it cannot create a target-size ladder or alter `T_selected`.

Foundation-model providers and large accelerator references are released as soon as their final preparation consumer completes. Derived file materialization and target-size candidate views run on CPU/I/O resources unless their current owner explicitly admits an accelerator task. Heavy caches are restored lazily only when a validated artifact is needed.

Candidate execution and continuation

The target-size screen executes only the cells authorized by the reducer's funnel:

```
qualified candidates
  -> coarse n1/M1
  -> at most four short n2/M2 continuations
```

```
  -> two final n3/M3 continuations
  -> one selected size or typed scientific failure
```

Each `(candidate size, optimizer seed)` cell runs through the accepted TRAIN2 runtime and current EVAL2 owner. A bounded numerical fake may sit below the accepted MACE seam in tests; configuration resolution, target authorities, materialization, checkpoint/provider authentication, persistence, and reducer publication remain production code.

At a fidelity boundary, continuation restores model, optimizer, EMA, LR, and Python/NumPy/Torch CPU/CUDA RNG state. It does not restart from the foundation or substitute an earlier checkpoint. Atomic content-addressed publication means an interrupted boundary either has a complete authenticated endpoint or has no current endpoint. The execution head is reconciled before new work is scheduled, and compare-and-set adoption prevents two workers from becoming current simultaneously.

Eliminated candidates receive no later ordinary-production authorization. Exhaustive full-fidelity training of every configured size is a separate algorithm/decision-preservation qualification, not a default campaign artifact generator.

Deterministic resource-bounded work queue

CPU-heavy independent tasks use a shared queue with explicit CPU, memory, and I/O ownership. Its responsibilities are to:

- bound executing, ready, in-flight, and buffered work;
- reserve persistent memory before admitting temporaries;
- propagate deterministic task identities and exceptions;
- permit arbitrary completion order where scientific order is irrelevant;
- restore canonical reduction and commit order where FP64 arithmetic or record order is authoritative;
- expose progress and resource telemetry without placing telemetry in scientific identity.

Submission may run ahead to hide hand-off latency, but simultaneous execution remains within the declared resource scope. On NUMA systems, node-local queues, affinity, local stealing, and bounded cross-node stealing are valid execution extensions after measurement; they cannot change canonical membership or reduction order.

Staged evaluation and provider lifetime

Current staged evaluation uses bounded CPU preparation, one admitted accelerator owner when applicable, and bounded CPU finalization. The parent execution owner enumerates the authenticated endpoints, workers perform only their assigned preparation/inference/finalization, and the parent validates run, checkpoint, selected-binding, prediction, and metric identities before durable publication. Fresh and cache-backed endpoints converge through the same parent validation.

Provider scopes are explicit and non-overlapping:

```
candidate provider acquire -> candidate inference/replay consumers
  -> candidate close in exception-safe cleanup
  -> foundation or next provider acquire only after closure
  -> foundation close in exception-safe cleanup
```

Post-selection CV and final-production providers are likewise closed at their owner boundary, including failure paths. A replay cache stores scalar/content evidence, not a live provider. Garbage-collection timing or allocator cleanup does not replace provider retirement.

A worker-private provider shell may be reused only when checkpoint bytes, model class, state keys/shapes/dtype, weight-independent runtime architecture, geometry workload, device, and backend policy all authenticate as compatible. Weight-dependent calculator state is invalidated on replacement. Corruption or authority mismatch is fatal rather than a fallback to an unqualified shell.

One authority per semantic input

The bounded execution representation is:

```
one canonical frame/feature authority
one neutral statistical substrate
one P_train/M3 split and pi_train/pi_eval
one common preparation
prefix views for candidate rungs
training and CV artifacts only for authorized work
```

Memory/storage must not scale as one product-sized descriptor, graph, or membership copy per target-size rung. Descriptor shards, fixed-file views, replay indexes, and frame caches are reconstructible only when their content and recipe identities authenticate. A cache hit is never a substitute for the selected binding or another scientific authority.

Memory, storage, and scratch admission

Long stages account for

$$M_{\text{stage}} = M_{\text{persistent}} + M_{\text{inflight}} + M_{\text{buffered}} + M_{\text{sparse}} + M_{\text{result}} + M_{\text{scratch}}.$$

New work is admitted only when CPU, RAM, accelerator, disk, and scratch reservations fit the stage plan. The live ledger is authoritative: a prospective target-size or evaluation reservation replaces only the exact modeled reservation it supersedes and preserves all other live owners. When retained growth is not bounded, sequencing is conservative rather than relying on an optimistic projection.

Large reconstructible arrays may use mmap/file-backed persistence. Atomic publish-or-validate-winner rules protect concurrent fixed-file and materialized cache creation. Stale, corrupt, or mismatched caches are rebuilt; they are not silently accepted as evidence.

Persistent campaign state uses a compact SQLite store, append-only event history where needed, content-addressed files for large payloads, and completion records written only after required artifacts are durable. A restart distinguishes complete, incomplete, stale, corrupt, and superseded state and re-authenticates currentness before reuse. Cleanup removes only known campaign-owned reconstructible state and preserves external inputs, selected scientific records, restart checkpoints, and diagnostics needed for recovery.

GPU/VRAM and host admission

GPU jobs are admitted against explicit device availability, free memory, and configured budget evidence. A one-job calibration establishes whether the applicable serial workload is viable; it does not by itself authorize parallel expansion. Soft utilization and fractional-VRAM envelopes regulate additional jobs, while a hard live-VRAM guard protects against OOM. Missing telemetry at calibration startup selects conservative serial execution when the device is otherwise usable; it does not create parallel evidence.

An execution controller may lower concurrency after measured resource pressure, but it cannot change scientific batch/exposure semantics, precision policy, checkpoint evidence, or target/replay membership to fit memory. OOM recovery is valid only when the retry is protocol-equivalent and the changed parameter is non-semantic.

Replay indexing and bounded parsing

The selected replay source remains external scientific authority. A reconstructible index may store source-byte identity, frame offsets/lengths, atom counts, and source-order geometry identity for sparse monitor access and bounded chunk parsing. Source mutation or index corruption causes safe reconstruction. Parser concurrency is added only when representative measurement shows benefit and exact replay bytes/identities remain unchanged.

Progress and observability

Every long-running stage exposes scientific progress and executor state:

1. completed/total work and percent where meaningful;
2. elapsed time and ETA when estimable;
3. throughput with an explicit stable unit;
4. active, pending, or buffered work;
5. resource pressure or the current hot item where relevant.

Heartbeats are emitted during long periods without task completion. ETA is based on globally committed work. User-facing elapsed and known ETA use fixed `HH:MM:SS`; unavailable ETA is `--:--:--`. Presentation state never enters scientific digests.

Performance qualification boundary

Performance changes are compared on representative work with equivalent scientific inputs and runtime conditions. Evidence records wall/CPU time, throughput, RSS/VRAM, scratch/storage, queue/backpressure, and output digests when material. A speedup obtained by changing precision, evidence population, ordering, or output is not a conforming optimization.

Target-machine GPU and long real-production qualification remain separate from P6 functional closure. They require their own supported hardware, workload, backend, and acceptance evidence; the current campaign does not infer those results from CPU or bounded numerical tests.

Part VII - Ownership and extension boundaries

One current-generation authority model

The current campaign has one semantic generation. A record, policy, or artifact is either authenticated for that generation or unsupported. Historical selector, repair, migration, and campaign-generation formats are not alternate current execution paths.

Architecture owns durable structure and scientific/algorithmic invariants. Specifications own exact schemas, policy values, tolerances, failure codes, and module-local behavior. Workplans coordinate proposed transitions and never become product authority merely because implementation follows them.

The core authority chain is:

```
source evidence and labels
    -> eligibility / conditions / evidence roles
    -> neutral statistical substrate and protected relations
    -> one P_train / M3 split
    -> one pi_train and nested pi_eval ladder M1 subset M2 subset M3
    -> one common target-size preparation
    -> paired optimizer-seed screen
    -> target-size reducer
    -> N_selected and exact global T_selected
    -> post-selection cross-validation on exactly T_selected
    -> fresh final production on the complete T_selected
    -> currentness-fenced publication
```

There is no branch to a second membership selector, a per-domain target-size map, a generated-size rescue, or downstream-evidence-driven fallback. Derived state from an unsupported generation is rejected before semantic reuse and is quarantined/reprepared rather than translated.

Scientific decision ownership

Decision or product	Sole current owner	Consumes	Emits	Explicitly does not own
source/label identity	DATA2-family contracts	immutable source material	normalized labelled-record identity	partition, selection, training
conditions/eligibility	DATA3-family contracts	source records	eligible frames and conditions	evidence-role assignment
raw features/events	DATA4-family contracts	eligible evidence and provider declarations	partition-independent raw evidence	fitted metrics or membership
evidence roles and protected relations	DATA5 partition contracts	cohorts, independence evidence, purge rules	neutral roles (development/monitor/calibration/locked) and split exclusions	post-selection CV folds or target ranking
descriptor/difficulty inputs	DATA6 contracts	authorized evidence and frozen foundation model	raw/blinded descriptors and predictions	target membership
common fitted preparation	current P3 preparation owner	neutral substrate and frozen method inputs	transforms, metrics, E0, objective/weights, difficulty inputs	membership or target size
target-size split and orders	current target-size experiment owner	frame authority, neutral substrate, configured policy	`P_train`/`M3`, `pi_train`, `pi_eval`, `M1`/`M2`/`M3`	method acceptance
common target-size preparation	`TargetSizeCommonPreparation`	`P_train` and foundation/training protocol	one shared preparation identity	per-size or per-seed scientific variation
scientific target size	one target-size reducer	paired target-side screen evidence	`N_selected` or typed scientific failure	monitor cardinality and CV evidence
current selected set	`CampaignStore` terminal projection	authenticated reducer state and `pi_train`	exact `N_selected`/`T_selected` binding	re-deciding size
post-selection method acceptance	post-selection CV owner	exactly `T_selected`, protected relations, `K >= 2`, CV seeds	all-required-fold target-only verdict	changing `N_selected`
fresh final production	final-production owner	accepted method, complete `T_selected`, required final seeds	complete executed run evidence / model artifacts	target-size or CV authority (publication is P7)
target monitor	current monitor policy	authorized development role	deterministic monitor	target membership
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

A narrow specification may refine a row's realization, but it cannot create a second semantic owner for the same decision.

Fitted preparation boundary

The current fitted-preparation owner may publish:

- heterogeneous feature transforms and metrics;
- foundation predictions and training-domain difficulty evidence;
- atomic-reference/E0 fits;
- training objective and configuration/property weight records;
- condition, provenance, event, environment, and diversity inputs;
- immutable identities linking products to their authorized inputs.

These are inputs to the one canonical order. They are not an independent quota, FPS, membership, target-size, or CV selector. A fold-local transform is allowed only after selection and only when the post-selection CV owner binds it to that fold's training partition; it cannot change global `T_selected`.

This boundary preserves useful fitted/statistical information without reintroducing a domain-specific target-size authority. Materialization and export records describe consumer views and remain downstream of the selected binding.

Target-size authority

Let (N_{}=|P_{}|). The candidate ladder is a configured contiguous power range,

$$\mathcal{N}_0 = \{2^p : p_{\min} \le p \le p_{\max}\},$$

with materializable population

$$\mathcal{N}_M = \{N \in \mathcal{N}_0 : N \le N_{\text{available}}\}.$$

The qualified population is the subset admitted by the current target-size policy. The selected size must be a qualified member. There is no hidden scientific ceiling beyond the configured policy and available population.

Every candidate is an exact prefix:

$$T_N = \pi_{\text{train}}[: N].$$

Thus frame membership is global, candidate sets are nested, and `N_selected`/`T_selected` are frozen together. Increasing `N` only adds frames; a pass/fail/pass result under a monotone prefix policy is an invariant failure, not a reason to choose a different order.

The reducer consumes only authorized target-side development/model-selection evidence. Replay metrics, post-selection CV, calibration, physical-observable, and locked-test evidence cannot rank, reject, or tie-break a target size. Fewer than three qualified sizes is a typed failure. A configured ceiling that remains materially superior at the final comparison produces `nonconverged_at_configured_ceiling`; no unconfigured rescue size is invented.

Post-selection ownership

The current post-selection graph is:

```
current selected binding
    -> shared method identity
    -> CV policy / final-production policy
    -> CV plan / final-production plan
    -> fold/final evidence
```

Cross-validation uses exactly `T_selected`, preserves P1 protected relations, requires every configured fold and seed, and accepts or rejects the method. It cannot alter `N_selected`. Final production starts fresh from the accepted foundation and trains the complete selected set under `[training].max_num_epochs`; it cannot continue a screen or CV run.

Every current consumer re-resolves the selected binding and current campaign revision before exposing a descendant. A stale caller-held object, checkpoint, or provider cannot become current. Publication rechecks currentness in the same transaction that would install a current pointer.

Public orchestration and storage boundary

The public scientific lifecycle is exactly:

```
init -> doctor -> prepare -> select-target-size -> cross-validate -> train-production
```

`prepare` builds the neutral/current substrate and common preparation but selects nothing. `select-target-size` owns candidate training and the reducer. `cross-validate` owns selected-only method acceptance. `train-production` owns fresh final publication. `status` and `advance` project these same owners; they do not create another state machine. `storage` is orthogonal and manages reconstructible artifacts without advancing scientific lifecycle.

Downstream qualification boundary

Deployment parity, physical PES/relaxation/dynamics validation, uncertainty calibration, and locked testing remain product obligations, but they are not current P6 campaign owners. A future successor may consume a current final publication through an explicit downstream contract. It may reject that publication, but it may not select another seed/checkpoint/member from downstream evidence and may not feed back into target-size or method authority.

Physical numerical algorithms remain owned by their analysis modules. A downstream recipe must bind matched collection/frame identity, runtime and capability identity, analysis-owned results, and a declared statistical role. P6 does not claim downstream implementation or qualification merely because its current final-production record is available.

Unsupported generations and compatibility

Current loaders do not semantically read or migrate obsolete target-size derived state. The narrow cutover detector may inspect record names or minimal generation metadata solely to reject before candidate/checkpoint reuse. It may quarantine the opaque record under a namespace no current loader reads, but it cannot decode, reconstruct, normalize, or bind that payload into current authority.

Independent lower-level source/frame/content caches may be reused only after their current owners revalidate source bytes, recipe, lineage, and integrity. Compatibility readers for non-target product responsibilities remain read-only and non-authoritative. Historical schemas and rationale belong under `docs/history/mlff/`; their presence does not create a current API.

Extension boundaries and summary

A future extension may enrich neutral feature/preparation inputs, extend the canonical `pi_train` policy, add an explicitly qualified screen metric, or change execution representation only when one-owner direction, exactness, protected evidence roles, and currentness remain intact. A new campaign generation is not entitled to automatic migration support.

The durable rules are:

1. independent evidence remains independent;
2. fitted preparation and target membership are separate authorities;

3. one canonical order and one common preparation define every candidate;
4. `N_selected` and exact global `T_selected` are frozen together;
5. the reducer is the sole target-size authority; post-selection cross-validation accepts the method and can never re-choose the size;
6. final production is fresh full-selected-set training;
7. execution/cache/provider choices cannot change scientific semantics;
8. unsupported derived state is rejected before reuse rather than migrated;
9. downstream qualification cannot become a P6 selection fallback.

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